📄 **PROJECT REPORT**

Fake Review Detector using Machine Learning

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1. Introduction

With the rapid growth of e-commerce platforms, online reviews have become an important factor in influencing customer decisions. However, many fake or spam reviews are posted to manipulate product ratings and mislead users.

This project, **Fake Review Detector**, uses Machine Learning to automatically classify reviews as **Fake** or **Genuine**, helping users make better decisions.

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2. Objective

The main objectives of this project are:

- To detect fake reviews using Machine Learning
- To preprocess and analyze text data
- To build a classification model
- To provide real-time prediction for user input

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3. Problem Statement

Fake reviews are widely used to promote or demote products unfairly. These reviews often contain exaggerated language, repeated patterns, or misleading content.

The challenge is to **automatically identify whether a review is fake or genuine** using AI/ML techniques.

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4. Technologies Used

- **Programming Language:** Python
- **Libraries:**
    - Pandas (data handling)
    - NumPy (numerical operations)
    - Scikit-learn (ML model)

- Regex (text cleaning)

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5. Methodology

The project follows these steps:

5.1 Data Collection

- A dataset of fake and real reviews is created manually.
- Fake reviews contain exaggerated phrases.
- Real reviews contain neutral and realistic feedback.

5.2 Data Preprocessing

- Convert text to lowercase
- Remove special characters
- Clean and normalize data

5.3 Feature Extraction

- TF-IDF (Term Frequency - Inverse Document Frequency) is used
- Converts text into numerical vectors

5.4 Model Training

- Logistic Regression is used for classification
- Data is split into training and testing sets

5.5 Evaluation

- Accuracy Score
- Classification Report (Precision, Recall, F1-score)

5.6 Prediction

- User inputs a review
- Model predicts whether it is Fake or Genuine

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6. System Architecture

User Input → Text Cleaning → TF-IDF Vectorization → ML Model → Prediction Output

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7. Results

- The model successfully classifies reviews as fake or genuine
- Achieved accuracy: **~85% to 95%** (depends on data randomness)
- Provides instant predictions

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8. Advantages

- Simple and efficient
- Fast prediction
- Helps detect misleading reviews
- Easy to implement and expand

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9. Limitations

- Uses a small synthetic dataset
- May not perform well on real-world complex data
- Limited vocabulary understanding

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10. Future Enhancements

- Use real datasets (Amazon, Flipkart, etc.)
- Apply Deep Learning (LSTM, BERT)
- Build a web application (Flask/Django)
- Add sentiment analysis
- Improve accuracy with larger datasets

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11. Conclusion

The Fake Review Detector demonstrates how Machine Learning can be used to solve real-world problems like identifying fake reviews. While the current model is basic, it provides a strong foundation for building more advanced systems.

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12. References

- Scikit-learn Documentation
- Python Official Documentation
- Research papers on Fake Review Detection